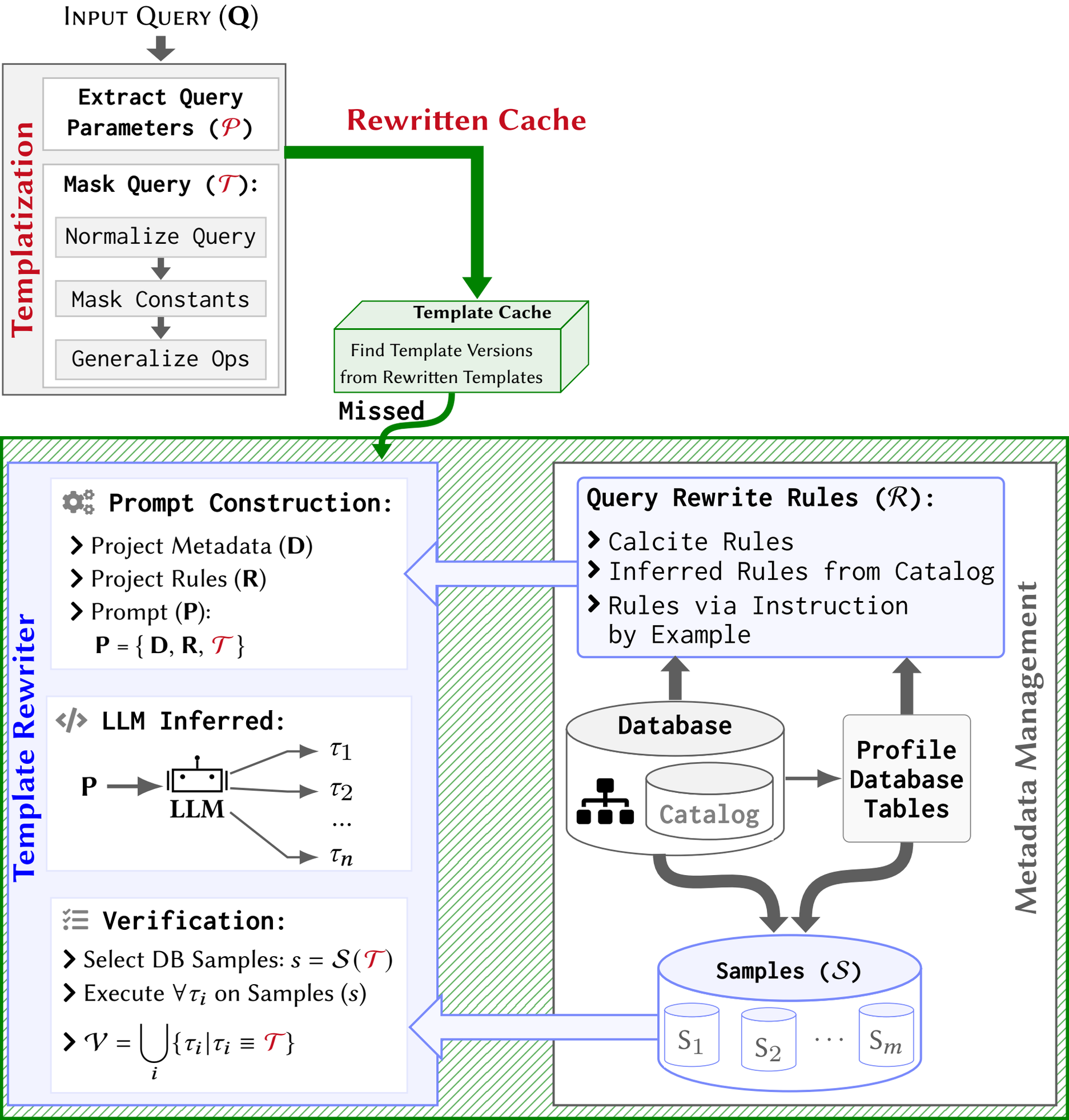

Input Query (Q)
Templatization
Extract Query Parameters (P)
Mask Query (T):
Normalize Query
Mask Constants
Generalize Ops
Rewritten Cache
Template Cache
Find Template Versions from Rewritten Templates
Missed
Template Rewriter
Prompt Construction:
Project Metadata (D)
Project Rules (R)
Prompt (P):
P = { D, R, T }
LLM Inferred:
P
LLM
τ1
τ2
...
τn
Verification:
Select DB Samples: s = S(T)
Execute ∀τi on Samples (s)
V = ⋃i {τi | τi ≡ T}
Query Rewrite Rules (R):
Calcite Rules
Inferred Rules from Catalog
Rules via Instruction by Example
Database
Catalog
Profile Database Tables
Samples (S)
S1
S2
...
Sm
Metadata Management